INCH-POUND
MIL-DTL-22052E(SH)
18 July 2018
SUPERSEDING
MIL-V-22052D(SH)
20 March 1978

DETAIL SPECIFICATION

VALVES, STOP AND STOP CHECK, GLOBE, ANGLE, AND Y PATTERN, CAST OR FORGED CARBON OR ALLOY STEEL, OUTSIDE SCREW AND YOKE (SIZES 2½ INCHES UP TO 16 INCHES)

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers stop and stop check; globe, angle, and Y pattern; cast or forged carbon or alloy steel; outside screw and yoke valves in sizes 2½ inches up to 16 inches.

1.2 Classification. Valves are classified in accordance with the following compositions (see table I) and ratings (see 1.2.1, 1.2.2, and 1.2.3), as specified (see 6.2).

1.2.1 Type. Valve are of the following types:

Type	PIN Code
Stop	S
Stop Check	C
Globe	G
Angle	A
Y Pattern	Y

1.2.2 Sizes. Valve sizes are of the following compositions:

Composition	PIN Code
Cro-Mo 2¼%, 1%	A
Cro-Mo 1¼%, ½%	B
Carbon Steel	D
Corrosion resistant steel	E

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1.2.3 Composition. Valves are fabricated in the following rating classes in accordance with ASME B16.34:

Rating Class	PIN Code
150	D015
300	D030
400	D040
600	S060
900	S090
1500	S150
2500	S250

1.2.4 Size. Valves are of the following sizes:

Size	PIN Code
2½	025
3	030
4	040
5	050
6	060
8	080
10	100
12	120
14	140
16	160

1.2.5 Potable water use. Valves are of the following potable water uses:

Category	PIN Code
Suitable	PW
Not Suitable	NS

1.3 Part or identifying number (PIN). The PINs for valves are created as follows:

M	22052	-	X	X	XXXX	/	XXX	XX
Prefix for Military Specification	Specification Number	-	Type (see 1.2.1)	Composition (see 1.2.2)	Rating (see 1.2.3)	/	Nominal Size (see 1.2.4)	Potable water use (see 1.2.5)
Example: M22052-GD030/050PW is for a Globe valve, Carbon Steel, Class 300, 5 inch, suitable for potable water use.								

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-1222	-	Studs, Bolts, Screws, and Nuts for Applications Where a High Degree of Reliability is Required; General Specification for
MIL-DTL-15024	-	Plates and Tags for Identification of Equipment, General Specification for
MIL-P-15024/5	-	Plate, Identification
MIL-R-17131	-	Rods and Powders, Welding, Surfacing
MIL-P-17416	-	Packing Material, Metallic, Flexible (Symbol 1430)
MIL-P-24396	-	Packing Material, Braided PTFE (Polytetrafluoroethylene)
MIL-P-24503	-	Packing Material, Graphitic, Corrugated Ribbon or Textured Tape and Preformed Ring
MIL-P-24583	-	Packing Material, Graphitic, Braided Yarn (Metric)
DOD-V-24657	-	Valve Actuator, Direct Coupled, Gear Driven Electrically Powered for Shipboard Propulsion and Auxiliary Systems (Metric)
MIL-DTL-24696	-	Gasket, Sheet, Non-Asbestos
MIL-G-24716	-	Gaskets, Metallic-Flexible Graphite, Spiral Wound
MIL-PRF-24790	-	Packing Material, Braided, Non-Asbestos

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited).
MIL-STD-798	-	Nondestructive Testing, Welding, Quality Control, Material Control, and Identification and Hi-Shock Test Requirements for Piping System Components for Naval Shipboard Use.
MIL-STD-2035	-	Nondestructive Testing Acceptance Criteria

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-470	-	Designing and Developing Maintainable Products and Systems, Volume I

(Copies of these documents are available online at http://quicksearch.dla.mil/.)

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

803-1385620 - Handwheels for Valves

803-2177525 - ¼ -2 BW & SW 900 LB ASME Valve

803-5001021 - Pressure Seal Ring-Standard and Oversize-Valve Pressure Class 600-1500

803-5184193 - Steel Globe Valves, 1/4 to 2 NPS, 1500 Class

(Copies of these documents are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of these documents may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

NAVSEA PUBLICATIONS

S9074-AR-GIB-010/278 - Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

T9074-AS-GIB-010/271 - Requirements for Nondestructive Testing Methods

(Copies of these documents are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering (hard copy) via the Naval Logistics Library at https://nll.ahf.nmci.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.1 - Unified Inch Screw Threads (UN and UNR Thread Form)

ASME B1.5 - Acme Screw Threads

ASME B16.10 - Face-to-Face and End-to-End Dimensions of Valves

ASME B16.20 - Metallic Gaskets for Pipe Flanges: Ring-Joint, Spiral-Wound, and Jacketed

ASME B16.34 - Valves Flanged, Threaded and Welding End

(Copies of these documents are available online at www.asme.org.)

ASTM INTERNATIONAL

ASTM A105/A105M - Standard Specification for Forgings, Carbon Steel, for Piping Components

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ASTM A106/A106M	-	Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service
ASTM A182/A182M	-	Standard Specification for Forged or Rolled Alloy-Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service
ASTM A193/A193M	-	Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service
ASTM A194/A194M	-	Standard Specification for Carbon and Alloy Steel Nuts for Bolts for High-Pressure and High-Temperature Service
ASTM A216/A216M	-	Standard Specification for Steel Castings, Carbon, Suitable for Fusion Welding, for High Temperature Service
ASTM A217/A217M	-	Standard Specification for Steel Castings, Martensitic Stainless and Alloy for Pressure Containing Parts Suitable for High-Temperature Service
ASTM A240/A240M	-	Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications
ASTM A312/A312M	-	Standard Specification for Seamless, Welded, and Heavily Cold Worked Austenitic Stainless Steel Pipes
ASTM A335/A335M	-	Standard Specification for Seamless Ferritic Alloy-Steel Pipe for High-Temperature Service.
ASTM A351/A351M	-	Standard Specification for Castings, Austenitic, for Pressure-Containing Parts
ASTM B700	-	Standard Specification for Electrodeposited Coatings of Silver for Engineering Use
ASTM F2168	-	Standard Specification for Packing Material, Graphitic, Corrugated Ribbon or Textured Tape, and Die-Formed Ring

(Copies of these documents are available online at www.astm.org.)

MANUFACTURERS STANDARDIZATION SOCIETY (MSS) OF THE VALVE AND FITTINGS INDUSTRY

MSS SP-25	-	Standard Marking System for Valves, Fittings, Flanges, and Unions
MSS SP-45	-	Bypass and Drain Connection
MSS SP-61	-	Pressure Testing of Valves
MSS SP-91	-	Guidelines for Manual Operation of Valves

(Copies of these documents are available online at http://msshq.org/Store/index.cfm.)

SAE INTERNATIONAL

SAE AMS3275	-	Sheet, Acrylonitrile Butadiene (NBR) Rubber and Non-Asbestos Fiber, Fuel and Oil Resistant
SAE AS15003	-	Fitting, Lubrication Hydraulic Surface Check, ⅛ Pipe Threads, Steel, Type III
SAE AS35411	-	Fittings, Lubrication

(Copies of these documents are available online at www.sae.org.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Standard and special rated valves furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3). Valves covered by this specification which have not been qualified shall undergo all qualification examinations of table VI before contract award for such valves.

3.2 Materials. Materials shall be as specified in table I. Materials for parts other than those listed in table I shall be selected to prevent galling, seizing, or excessive wear of operating parts. Cast iron and aluminum shall be used only for those parts where permitted in this specification. Magnesium alloys shall not be used. Clearances shall be adequate to prevent interferences due to thermal expansion.

TABLE I. List of materials.

Name of part	Material form	Composition A	Composition B	Composition D	Composition E
		Applicable documents			
Body, bonnet	Forgings or castings	ASTM A182/A182M, grade F 22 ASTM A217, grade WC 9	ASTM A182/A182M, grade F 11 class 2 and grade F 22 class 3 ASTM A217, grade WC 6	ASTM A105, ASTM A216, grade WCB [1/]	ASTM A182/A182M, grade F 304 or grade F 216 ASTM A351/A351M, grade CF3 or grade CF3M
Retaining ring and yoke	Forgings or castings	ASTM A182/A182M, grade F 22 or grade F 11 ASTM A105 ASTM A217, grade WC 9 or grade WC 6 ASTM A216, grade WCB			ASTM A182/A182M, grade F 304 or grade F 216 ASTM A351/A351M, grade CF3 or grade CF3M ASTM A240/A240M, grade 304 or grade 316
Studs for bonnets or bonnet clamps	Allow steel for high temperature bolting	ASTM A193, grade B 16	ASTM A193, grade B 16	ASTM A193, grade B 7 or grade B 16	ASTM A193/A193M, grade B 7
Nuts for bonnets or bonnet clamps	Heavy semi-finished hexagon carbon and alloy steel for high temperature bolting	ASTM A194, grade 4	ASTM A194, grade 4	ASTM A194, grade 2 H or grade 4	ASTM A194/A194M, grade B 7

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TABLE I. List of materials - Continued

Standard pressure seal rings and standard oversized pressure seal rings [2/]	Soft carbon steel, silver plated	803-2177525, 803-5001021 or 803-5184193 as applicable.
Alternate compression pressure seal rings [3/]	Graphite material	MIL-P-24503 or ASTM F2168
Gaskets (for flanged bonnets)	Spiral wound	MIL-G-24716, type I, class A or B
	Compressed gasket sheet	MIL-DTL-24696 type I (except where fuel resistance is necessary) SAE AMS3275 (where fuel resistance is necessary)
Packing	Packing material, graphitic, or polytetrafluoro ethylene	MIL-P-24583 (nonlubricated) service temperature not to exceed that of saturated steam MIL-P-24396 (PTFE) for service not to exceed 450 °F MIL-P-24503, superheated steam total temperature in excess of 350 °F
	Plastic, metallic or nonmetallic	MIL-P-17416 MIL-PRF-24790
Nuts, bolts, washers, bushings, liners	As per applicable drawing or, if not specified therein, materials used in construction of these parts other than specified above shall be in accordance with MIL-DTL-1222.	
Valve trim	(see table IV)	

NOTES:
1/ Bonnet material for composition D valves may also be grade WC6 of ASTM A217.
2/ Oversized seal rings may be used in the repair of pressure seal bonnets, (i.e., as shown on 803-5001021).
3/ Graphite seal rings may be used in replacement of standard steel ones, with capped upper and lower seals to preclude packing extrusion.

3.3 Design. Unless otherwise specified herein, valves, valve parts, and design features and parameters shall be in accordance with ASME B16.34. Minimum flow coefficients (C_v) shall be as specified (see 6.2).

3.3.1 Pressure-temperature ratings. The design and pressure-temperature rating shall be in accordance with ASME B16.34, except the maximum allowable temperature for composition D shall be 775 °F. The pressure seal valve shall be constructed in such a manner as to permit repair by the use of oversize rings. The detail design of the oversize seal ring shall be as shown on 803-5001021.

3.3.2 End connections. Valve end connections shall withstand the forces and moments imposed by the connecting pipe to which they are attached. For design purposes, the maximum value of the fiber stress in the connecting pipe produced by these forces and movements shall be considered to be equal to 0.2 percent effect yield stress of the piping material at room temperature.

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3.3.3 Drains and bypasses. Drains and bypasses shall be in accordance with the requirements specified in 3.3.3.1 through 3.3.3.5. A drain shall consist of a nipple and drain valve. A bypass shall consist of connecting lines and a bypass valve. Installation shall be in accordance with MSS SP-45 except that welded connections shall be used rather than threaded.

3.3.3.1 Size of drains and bypasses. Standard drain and bypass sizes shall be as shown in table II. Nonstandard sizes shall be as specified (see 6.2).

TABLE II. Standard drain and bypass sizes.

Valve size (inches)	Size of bypass and drains (inches)
2½ and 3	½
4, 5, 6, and 8	¾
10 and 12	1
14 and 16	1¼

3.3.3.2 Location. The location of drains and bypasses shall be specified by referencing the letter designation in ASME 16.34 of the desired bosses (see figure 1 and 6.2). Bosses shall be a sufficient distance away from the seating area to allow welding of replacement bypass lines without damage to the valve seat. When nonstandard locations are required, a drawing shall be specified and furnished by the procuring activity (see 6.2) indicating the desired location. When a location is not specified, the following standard locations shall be used:

a. Bypass – "A" to "B" of figure 1 (for globe and Y pattern valves), "E" to "F" of figure 1 (for angle valves).

b. Drain – "C" of figure 1 (for globe and Y pattern valves).

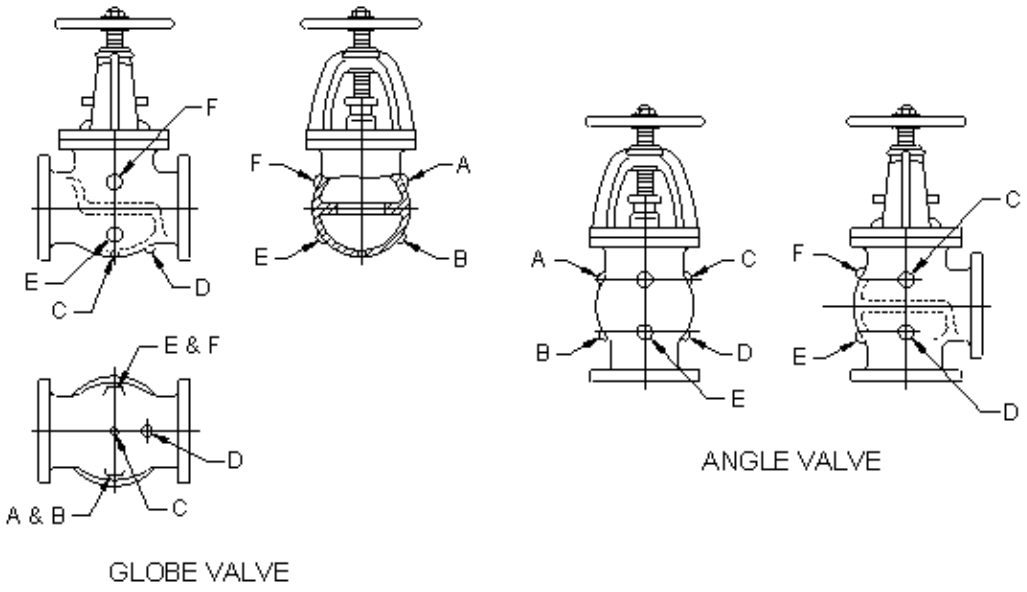


FIGURE 1. Designating boss locations for draining and bypasses.

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3.3.3.3 Root connections. Drain and bypass line connections shall be welded to the boss in the valve body in accordance with S9074-AR-GIB-010/278. Materials for these lines shall be as follows:

a. Composition A – ASTM A335, grade P22.

b. Composition B – ASTM A335, grade P11.

c. Composition D – ASTM A106, grade B.

d. Composition E – ASTM A312/A312M, grade 304 or grade 316.

3.3.3.4 Bosses. When specified (see 6.2), bosses shall be required for valves of class 600 and above in sizes 2½ inches and larger and valves less than class 600 in sizes 4 inches and larger when these valves are furnished without a bypass and drain. When specified (see 6.2), the valve bodies shall be provided with bosses as shown on "A", "B", and "C" of figure 1 for globe and Y pattern valves and as shown on "E" and "F" of figure 1 for angle valves.

3.3.3.5 Drain and bypass valves. Drain and bypass valves shall be in accordance with 803-2177525 or 803-5184193 as determined by pressure, temperature, and material requirements of the application and shall have welding ends.

3.3.4 Body pattern. The body pattern (in-line, angle, or Y pattern) shall be as specified (see 6.2).

3.3.5 Port arrangement. Unless otherwise specified (see 6.2), the port arrangement on globe valves shall be in-line.

3.3.6 Weights and center of gravity. Handwheel operated, welded and globe, angle, and Y pattern valves shall not exceed the maximum weights listed in table III. Weights are based on valves with welded ends and do not include weights of drains, bypasses, operators, etc. Valves in classes and sizes not listed in table III do not have weight requirements. Valves with weights of 100 pounds or greater shall be marked with the locations of the vertical, longitudinal, and transverse center of gravity.

TABLE III. Maximum weights of welded end valves.

Valve size (inches)	Valve weights (pounds)						
	Class 150	Class 300	Class 400	Class 600	Class 900	Class 1500	Class 2500
2½	64	80	--	100	--	165	450
3	64	105	--	155	223	220	670
4	108	160	240	230	414	530	800
5	136	250	355	485	515	800	1775
6	207	385	730	600	770	820	2290
8	307	650	1100	800	840	1500	3500
10	500	1820	1630	1620	1800	2500	5000
12	730	1610	2090	2250	2500	3000	--
14	--	2375	--	3510	3300	3950	--
16	--	3000	--	--	5000	5600	--

3.3.7 Face-to-face and end-to-end dimensions. Face-to-face and end-to-end dimensions shall be in accordance with ASME B16.10.

3.3.8 Shock and vibration. Valves shall withstand the shock requirements of MIL-STD-798 and the vibration requirements of MIL-STD-167-1, type I (see 4.5.4). Valves with gear or power actuators attached shall withstand these requirements.

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3.3.9 Bonnet and yoke construction.

3.3.9.1 Pressure seal and breech lock bonnets. Class 600, 900, 1500, and 2500 valves shall have pressure seal or breech lock bonnets.

3.3.9.1.1 Threads. When retaining rings are threaded, the threads shall be either chromium or nickel-plated.

3.3.9.1.2 Bonnet seal rings. Bonnet seal rings shall provide a seal either by plastic or elastic expansion. Where a seal is obtained by means of plastic expansion, the seal ring shall have a maximum Brinell hardness number (Bhn) of 100. Where a seal is obtained by means of elastic expansion, the seal ring shall have a maximum Bhn of 140. Seal rings shall be silver-plated in accordance with ASTM B700 grade S with thickness of 0.002 to 0.003 inch. A supply list detailing oversized seal rings for repair purposes shall be as specified (see 6.2). The bonnet seal ring region of the valve body shall be inlayed with corrosion-resistant steel in accordance with S9074-AR-GIB-010/278. Graphite seal rings may be used in replacement of standard steel ones with capped upper and lower seals to preclude packing extrusion.

3.3.9.2 Flanged bonnets. Classes 150, 300, and 400 valves shall have bonnets with the joint faces of the male and female type or small tongue and groove type, with spiral wound gaskets in accordance with MIL-G-24716 or other gasket material meeting or exceeding MIL-G-24716 requirements.

3.3.9.2.1 Bearing surfaces. Bearing surfaces of nuts and bolt heads and their respective mating surfaces on the valves shall be machine finished.

3.3.9.2.2 Fasteners. Fasteners not otherwise specified herein shall be in accordance with MIL-DTL-1222 and the following:

a. Through-bolt type shall be bolt studs with studs threaded the entire length.

b. Studs of the single nut type shall have a 1¼ diameters fully formed thread engagement; lead vanishing threads shall not be considered.

c. Cap screws shall have a 1 inch diameter fully formed thread engagement; lead vanishing threads shall not be considered.

3.3.9.3 Yoke construction. Valves shall be of outside screw and yoke design.

3.3.10 Stuffing boxes and glands.

3.3.10.1 Stuffing boxes. Stuffing boxes shall be of a depth to accommodate packing that will ensure a pressure tight seal at the rated pressure and temperature of the valve, and to ensure a positive engagement between the stuffing box and gland. Depth shall be the minimum required to achieve a seal. When graphitic material seals are used, no more than four anti-extrusion rings shall be allowed.

3.3.10.2 Glands. Stuffing box glands shall be accrued and adjusted by a bolting arrangement which by adjustment will insure tightness of the stuffing box under all operating conditions. Bearing surface between the gland follower and the gland flange shall be spherically finished.

3.3.11 Trim.

3.3.11.1 Stem. Valve stems shall have Acme type threads in accordance with ASME B1.5 for valve operation and Unified National Standard threads for handwheel attachment in accordance with ASME B1.1. Unless otherwise specified (see 6.2), the valve shall have a rising stem construction.

3.3.11.1.1 Stem attachment. The stem of a stop check valve shall not be attached to the disc. The end of the stem shall be constructed to serve as a disc guide throughout the full travel of the disc.

3.3.11.2 Disc construction.

3.3.11.2.1 Disc attachment. Attachment of the disc to the stem of the stop valve shall be of swivel construction. The use of balls or slip-on type is prohibited in the design of the swivel feature in the stem to disc attachment. Furthermore, the combined design features of the disc and the swivel attachment shall be such that the disc will not spin when subject to the normal flow conditions of the valve.

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3.3.11.2.2 Disc design. The disc of stop check valves shall be of the piston guide and dash pot design. The disc shall fit into the body in such a manner as to serve as a guide during the full travel of the disc. The dash pot effect shall provide an effective cushion for the disc and prevent vibration and hammer at low velocities or pulsating loads. The disc shall be free of protrusions that could cause spinning of the disc.

3.3.11.2.3 Stop valve discs. For stop valves, provisions shall be made to prevent galling between the end of the stem and the disc. This may be accomplished by the insertion of a hardened washer (minimum 485 Brinell) between the stem and disc, by a pad of stellite inlaid on the inside of the disc at the point of contact or on the bottom of the stem, or a cast Stellite disc.

3.3.11.3 Main seat construction. All valves should have sufficient seat material to initially lap the seats. When possible, valves should have sufficient material for relapping as expected over the service life of the valve. Main seat shall be:

a. Integral.

b. Threaded. When threaded construction is used in valves intended for steam service, the seat ring shall be seal welded circumferentially so as to prevent leakage past the seat rings.

c. Welded-in.

3.3.11.4 Back seat. Valves shall have a positive back seat. Class 600 and higher valves shall have hard facing (HF) back seats (see 3.3.11.6) that shall be the same material as the main seat, using seat material as specified in table IV. (Note that the term “hard facing” is sometimes referred to as “hard surfacing” in other welding publications, including S9074-AR-GIB-010/278.)

3.3.11.5 Valve trim material. Unless otherwise specified (see 6.2), valve trim materials shall be in accordance with table IV.

TABLE IV. Valve trim materials.

Service	Valve trim symbols [1/]			
	Stem	Disc [2/]	Seat [2/]	Temperature limitation (°F)
Steam [3/]	Cr 13 [5/]	HF [3/]	HF	1050
Water [4/]	Cr 13	M1 – CU	Cr 13	750
	Cr 13	HF	HF	1050
Oil	Cr 13	Ni – Cu	Cr 13	500
	Cr 13	Cr 13	Cr 13	1000
	Cr 13	HF	HF	1050

NOTES:
1/ HF may be substituted for other seats and discs under all steam temperature and service conditions.
2/ Where unlike seats and discs are used, either the seat or the disc may be Cr 13 or Ni-Cu.
3/ Steam valves shall have HF seats and discs.
4/ Water valves pressure class 600 and higher shall have HF seats and discs.
5/ Cr 13 trim is equivalent to American Petroleum Institute specifications valve trim #8.

3.3.11.6 Hard facing. Welding materials shall be in accordance with type MIL-CoCr-A of MIL-R-17131. The minimum finished thickness of HF seating surfaces shall be 3/32 inch.

3.3.12 Valve handwheels.

3.3.12.1 Rotation. Valves shall close by clockwise rotation of the handwheel when facing the handwheel.

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3.3.12.2 Material and design. Handwheels 11 inches in diameter and smaller shall be manufactured from steel, ductile iron, malleable iron, or aluminum. Handwheels 12 inches in diameter and larger shall be aluminum alloy or cast steel in accordance with the general dimensions of 803-1385620. Steam handwheel connection shall be in accordance with 803-1385620.

3.3.12.3 Size. Handwheels shall be sized so that a tangential force less than or equal to the value specified in table V is required to be exerted on the rim of the handwheel to affect tight closure (see 4.5.5.2). Valve operating devices may be necessary to accomplish effective closure. These actuators shall comply with MSS SP-91.

TABLE V. Maximum allowable tangential force to seat valves based on valve handwheel size.

Hand-wheel diameter (inches)	Maximum allowable tangential force to seat valve (pounds)	Maximum allowable torque to seat valve (foot-pounds)	Maximum allowable tangential force to open valve (pounds)	Maximum allowable torque to open valve (foot-pounds)
2	90	8	40	3
3	98	12	40	5
4	106	18	40	7
5	112	23	40	8
6	118	30	40	10
7	121	35	40	12
8	124	41	40	13
9	127	48	40	15
10	130	54	40	17
11	133	61	40	18
12	135	68	40	20
14	138	81	40	23
16	141	94	40	27
18	144	110	40	30
21	147	131	40	35
24	150	150	40	40
27	150	169	40	45
30	150	188	40	50
36	150	225	40	60

3.3.12.4 Hammer-blow wheel. Class 600 valves, 4 inches and larger, and valves of class 1500, 3 inches and larger, shall be equipped with hammer-blow wheels unless a valve operator is specified.

3.3.12.5 Toggle operators. Toggle operators shall be of double toggle design and shall consist of eight equal length toggle arms with toggle arrangement being mounted on the valve yoke. Valve clamp rings shall not be a part of the valve yoke or toggle operator. Toggle design shall include a means of avoiding overstressing of the valve stem by providing an ample amount of stops to limit the stress applied to the valve stem. A means to ensure a constant seating lead on the valve seats when the valve is in the closed position shall be provided.

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3.3.12.6 Power operators. Valves shall be constructed to ensure a means of mounting power (electric, hydraulic, pneumatic, or other) actuators on the valves. Valves provided with electric actuators shall be in accordance with the requirements of MIL-DTL-24657. Specific closure times, if different from those of MIL-DTL-24657, shall be as specified (see 6.2).

3.3.13 Seat leakage. Unless otherwise specified (see 6.2), allowable seat leakage is leakage of water less than or equal to ⅓ liquid ounce (oz.) (10 cubic centimeters [cm^3]) per hour per inch of nominal pipe size (NPS) when tested in accordance with 4.5.5.2. The leakage rate through a closed check valve seat shall not exceed ⅘ oz. (25 cm^3) per hour per inch of NPS.

3.3.14 Lubrication. Yoke bushings 2½ inches and larger shall be equipped with a ⅛-inch type III lubricating fitting in accordance with SAE AS35411 and SAE AS15003.

3.3.15 Welding and fabrication.

3.3.15.1 Welding. Welding shall be in accordance with S9074-AR-GIB-010/278.

3.3.15.2 Fabrication. Fabricated assemblies shall be stress relieved as units prior to finish machining. Stress relieving shall be done in accordance with S9074-AR-GIB-010/278.

3.3.16 Maintainability. Valve design and construction shall ensure that assembly and disassembly of the valve can be accomplished without the need for special training or special tools. Special tools are defined as those tools not listed in the National Supply Catalog (copies of this catalog may be obtained by contacting the office of the Defense Contract Administration Service). Design and external configuration shall permit the use of a portable boring machine to repair the bonnet inlay of the valve with the valve in place in the piping system. When the intended end use of the valve is a shipboard system installation, master drawings shall be as specified (see 6.2 and 6.4).

3.4 Body markings and identification plates.

3.4.1 Body markings. Valve bodies shall have the class and manufacturer's name or trademark cast or forged integral with the valve body. Globe valves shall have a bridge wall marking in addition to the above markings. When necessary, metal stamping is permitted on the neck of the valve body or other similar surface not subjected to high stresses in service. In addition to the requirements herein, markings shall comply with MSS SP-25.

3.4.2 Identification plates. Identification plates made of corrosion-resistant steel or brass shall be permanently fastened to a part of the valve not subjected to working pressure, preferably the yoke in accordance with MIL-DTL-15024 and MIL-P-15024/5, and shall include the following data or space therefore:

a. Manufacturer's name or trademark.
b. Size of valve and class.
c. Stop check valve, if so constructed.
d. Body and bonnet material composition.
e. Valve trim identification (stem-disc-seat).
f. Manufacturer's identification number (optional).
g. Manufacturer's drawing number.
h. MIL-DTL-22052.
i. Component identification number, if known.
j. National stock number, if known.
k. C_v

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall be conducted at a laboratory acceptable to NAVSEA. Qualification inspection shall include the examinations and tests indicated in table VI.

TABLE VI. Examination and tests.

Characteristics	Requirement	Qualification	Conformance
Materials	3.2	4.4 and 4.5.1	--
Design	3.3	4.4	4.4
Flow coefficient (C_v)	3.3	4.5.3	--
Pressure-temperature rating	3.3.1	4.5.3	4.5.2
End connections	3.3.2	4.4	--
Drains and bypasses	3.3.3	4.4	--
Body pattern	3.3.4	4.4	--
Port arrangement	3.3.5	4.4	--
Weight and center of gravity	3.3.6	4.4	--
Face-to-face and end-to-end dimensions	3.3.7	4.4	4.4
Shock and vibration	3.3.8	4.5.4	--
Bonnet and yoke	3.3.9	4.4	--
Stuffing boxes and glands	3.3.10	4.6	--
Trim	3.3.11	4.4	--
Handwheels	3.3.12	4.4	4.4
Seat leakage	3.3.13	4.5.5	4.5.5
Lubrication	3.3.14	4.6	--
Welding and fabrication	3.3.15	4.5	--
Maintainability	3.3.16	4.6	--
Body markings and identification plates	3.4	4.4	4.4
Operating Torque	3.3.12.3	4.5.3.3c	--

4.2.1 Qualification inspection samples. Valves submitted for qualification inspection shall be of the 2½ inch size, special class 1500, composition B. Valves of the 2½ inch size, special class 1500, composition B, which have passed the qualification tests, will qualify valves in sizes 2½ inches and larger of all compositions, classes 600, 900, 1500, and 2500, provided variations in design throughout the various sizes and classes are demonstrated by analysis to be satisfactory. Each of the sample valves selected shall be examined in accordance with table VI and table VIII. Any valves in the sample containing one or more defects shall be rejected. If the number of defective valves in any sample exceeds the acceptable number for that sample, the lot represented by the sample shall be rejected.

4.3 Conformance inspection. Conformance inspection of lot samples shall include the examinations and tests indicated in table VI.

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4.3.1 Lot. A lot shall consist of valves of the same size, class, and composition offered for delivery at the same time.

4.3.2 Sample. A random sample of valves shall be selected from each lot offered to the Government in accordance with ANSI/ASQ Z1.4 at inspection level II and the acceptance quality limit (AQL) specified (see 6.2).

4.4 Visual and dimensional examination. Valves shall be visually and dimensionally examined to verify compliance with the requirements of this specification not involving tests, including weighing and establishment of the location of the center of gravity.

4.5 Tests.

4.5.1 Nondestructive testing (NDT). NDT shall be conducted as follows or if other service conditions are specified (see 6.2).

4.5.1.1 Radiographic inspection. Pressure containing castings of valves shall be 100 percent radiographically inspected when the service pressure or temperature will exceed the values listed in table VII. Inspection shall be in accordance with the following:

a. Radiographic inspection shall be in accordance with T9074-AS-GIB-010/271.

b. Radiographic acceptance shall be in accordance with S9074-AR-GIB-010/278.

c. Radiographic coverage shall be in accordance with the shaded areas shown on figure 2 and figure 3.

TABLE VII. Pressure or temperature criteria for NDT.

Service [1/]	**Gage pressure(lb/in^2)**	**Temperature (°F)**
Steam	225	500
Water	600	500
Oil	300	150

NOTE:
[1/] When valves are used in services other than those listed, the criteria for NDT shall be as specified (see 6.2).

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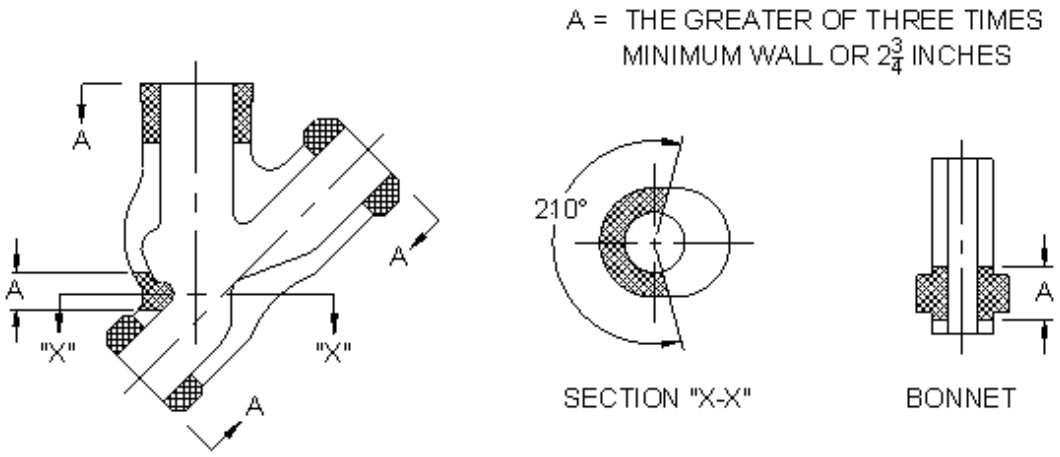


FIGURE 2. "Y" pattern globe body (pressure seal bonnet).

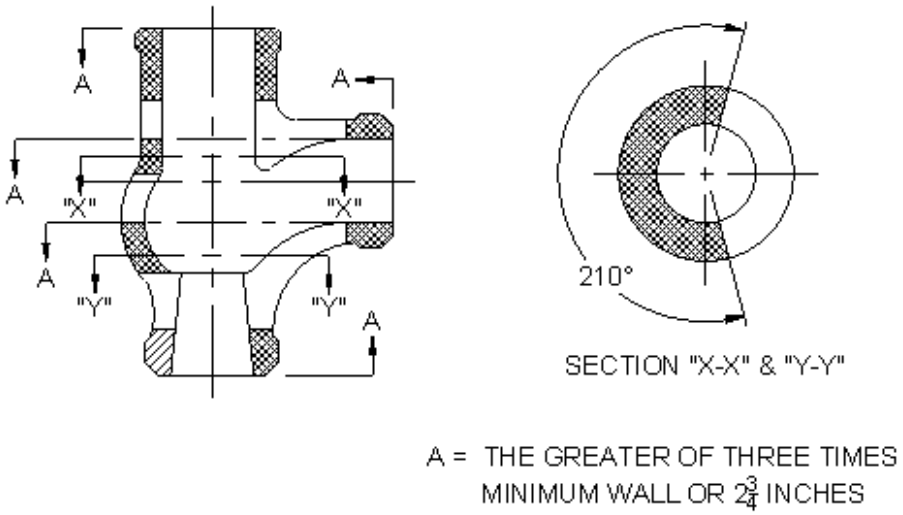


FIGURE 3. Angle body (pressure seal bonnet) bonnet same as "Y" pattern globe.

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4.5.1.2 Magnetic particle inspection. Pressure containing forging of valves shall be 100 percent magnetic particle inspected in accordance with T9074-AS-GIB-010/271 when the service pressure or temperature exceeds the values listed in table VII. Forgings shall be free from flaws such as seams, cracks, laps, porosity, scale, flakes, and all other defects detrimentally affecting the suitability for the service intended. Acceptance shall be in accordance with MIL-STD-2035.

4.5.1.3 NDT defects to be repaired. The following shall apply to those parts inspected in accordance with 4.5.1.1 and 4.5.1.2:

a. Defects less than 0.030 inch in depth do not need to be repaired provided the bottom of the defect is rounded and visible and the minimum wall thickness is maintained.

b. Defects greater than 0.030 inch in depth and less than 15 percent of the wall thickness shall be repaired by removing the defective material. This material shall be removed by drilling or grinding to a bottom radius of at least three times the depth of the defect. The depth of the finished repair shall be less than 15 percent of the undamaged wall thickness. Sharp corners shall be faired into the base metal. Welding is not required to affect a repair of this nature, provided the minimum wall thickness is maintained.

c. Defects greater than 15 percent of the wall thickness shall be repaired by removing the defective material and welding. The material shall be removed by grinding or drilling and then welded in accordance with 3.3.15. The crown of the weld shall be blended into the base metal.

4.5.1.4 Hard faced seating. After rough machining, hard faced seating surfaces shall be liquid penetrant inspected in accordance with T9074-AS-GIB-010/271 and shall be free of cracks or crack-like defects.

4.5.2 Cold tests. Cold tests shall be performed at 100 °F as follows:

a. Operation – 10 cycles.

b. Hydrostatic – shell and closure tests (see 4.5.5.1 and 4.5.5.2).

4.5.3 Hot tests. Hot tests shall be conducted with a gage pressure temperature as defined in 3.3.1 and shall be performed in the following sequence:

a. Thermal shock.

b. Operation – 200 cycles.

4.5.3.1 Thermal shock tests. Valves shall be thermal shocked 10 times by reducing the steam temperature from maximum rated temperature by a minimum of 35 percent in 30 seconds or less. Steam temperatures shall be measured as close to the valve as possible, preferably with the temperature sensing inside the valve. In addition to the steam temperature, a thermocouple shall be attached to the exterior of the valve in a zone of maximum temperature and minimum wall thickness. This thermocouple shall measure each thermal shock cycle to determine the temperature of the valve at the start of the cycle, the minimum temperature obtained during the cycle, the time required to reduce the valve to the minimum temperature, and the time required to restore the valve to the maximum temperature.

4.5.3.2 Operational tests. Steam shall be passed through the valve for a period of at least 48 hours. During this period, two securing trials shall be made to determine if the valve remains tight while and after cooling to ambient temperature and two warm-up trials shall be made to determine the torque required to open the valve which has been closed while cold. The valve shall be cycled a minimum of 200 times and the operating torque shall be recorded periodically during the cycling. The C_v shall be verified during this test.

4.5.3.3 Acceptance criteria. Acceptance criteria for the preceding tests shall be as follows:

a. Permanent deformation, misalignment, and functional impairments shall be cause for rejection.

b. Permissible seat leakage shall be as specified in 3.3.13.

c. No valve of operating torque shall exceed maximum allowable torque (see 3.3.12.3).

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4.5.4 Shock and vibration. Valves shall be subjected to the shock test in accordance with MIL-STD-798 to determine conformance to the requirement specified in 3.3.8 and the type I vibration test in accordance with MIL-STD-167-1. Valves shall be shock tested throughout, at full open positon, while hydrostatically pressurized to 100 percent of the 100 °F rating as specified in 1.2.3 and 3.3.1and on the inlet side at the full closed position. The testing medium shall be clean tap water with no additive other than traces of cutting oil at a temperature no greater than 150 °F and shall not be less than 40 °F. Momentary malfunction at the time of impact blow is permitted.

4.5.4.1 Acceptance criteria. Acceptance criteria for shock and vibration tests shall be as follows:

a. Permanent deformation, misalignment, and functional impairments shall be cause for rejection.

b. Permissible seat leakage shall be as specified in 3.3.14.

4.5.5 Hydrostatic tests. Valves shall be subjected to the tests specified in 4.5.5.1 for strength and porosity and the tests specified in 4.5.5.2 for tightness. Test shall be conducted in accordance with MSS SP-61; except that criteria herein take precedent over those of MSS SP-61. Water temperature shall not exceed 100 °F.

4.5.5.1 Shell test. Valves shall be given a shell test at a gage pressure no less than 1½ times the 100 °F rating and rounded off to the next higher 25 pounds per square inch (lb/in^2) increment. The test shall be made with water (which may contain a corrosion inhibitor), with kerosene, or with other suitable fluid, provided such fluid has viscosity not greater than that of water at a temperature not above 125 °F. Visually detectable leakage through pressure boundary walls is not acceptable. Test duration shall be not less than as follows:

Valve NPS (inches)	**Test time (seconds)**
2½ - 8	60
10 and larger	180

Tests shall be made with the valve in the partially open position. Leakage through the stem packing, not under pressure, shall not be cause for rejection.

4.5.5.2 Closure tests. Following the shell tests, valves shall be given a closure test. Each valve 10 NPS and larger, regardless of class, and each valve in the size range of 4 NPS through 8 NPS, Class 600 and higher shall be given a closure test using a fluid described in 4.5.5.1 at a pressure no less than 110 percent of the 100 °F pressure rating. Each valve 4 NPS through 8 NPS, less than class 600 and each valve less than 4 NPS, regardless of the pressure rating, shall be given either a fluid closure test at a pressure no less than 110 percent of the 100 °F pressure rating or a gas closure test at a minimum gage pressure of 80 lb/in^2. The test pressure shall be applied successively on each side of the closed valve. Duration of the closure test shall be the same as specified in 4.5.5.1. Permissible seat leakage shall be as specified in 3.3.13.

4.5.5.2.1 Closure test conduct. For globe and angle valves, the test pressure shall be applied across the closure number in the direction producing the most adverse seating condition using the operator or operating method intended for shipboard use. A stop check valve or other valve type designed to be a one-way valve, requires a closure test only in the appropriate direction.

4.5.5.2.2 Limited operating conditions. The closure test may be reduced to 110 percent of the maximum specified closed position differential pressure for those valves intended for operating conditions that have the pressure differential across the closure member limited to less than the 100 percent pressure rating of the valve design, or having closure members or actuating devices (direct, mechanical, fluid, or electrical) that would be subject to damage at high differential pressures.

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4.6 Examination after tests. After the tests of 4.5 have been completed, valves shall be disassembled and visually and dimensionally examined for damage and wear. Any valve or sample lot meeting the defect criteria of table VIII shall be rejected unless repairable as specified in 4.5.1.3. Disassembly and reassembly shall be performed to determine the practicability of maintaining a valve of this design for shipboard use (see 3.3.13). The maintainability demonstration shall be conducted by removing and replacing the pressure seal ring or seal welded flexible seal ring, disc, and stem. The maintenance procedures, removal of parts for repair, and the time required for accomplishment shall be verified during qualification testing. The demonstration procedures in MIL-HDBK-470 may be used. Particular emphasis shall be directed toward the possibility of loss of small parts by the maintenance crew.

TABLE VIII. Classification of defects.

Categories	Critical Defects
01	Composition and rating not as specified (see 1.2).
02	Material not as specified (see 3.2).
03	Design not as specified (see 3.3).
04	C_v not as specified (see 3.3).
05	Pressure temperature rating not as specified (see 3.3.1).
06	End connections not as specified (see 3.3.2).
07	Drains and by-passes not as specified (see 3.3.3).
08	Body pattern not as specified (see 3.3.4).
09	Port arrangement not as specified (see 3.3.5).
10	Weight and center of gravity not as specified (see 3.3.6).
11	Face-to-face and end-to-end dimensions not as specified (see 3.3.7).
12	Shock and vibration not as specified (see 3.3.8).
13	Bonnet and yoke not as specified (see 3.3.9).
14	Stuffing box and gland not as specified (see 3.3.10).
15	Trim not as specified (see 3.3.11).
16	Handwheels not as specified (see 3.3.12).
17	Seat leakage not as specified (see 3.3.13).
18	Lubrication not as specified (see 3.3.14).
19	Welding and fabrication not as specified (see 3.3.15).
20	Maintainability not as specified (see 3.3.16).
21	Body markings and identification plates not as specified (see 3.4).

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Steel stop and stop check globe, angle, and Y pattern valves in sizes 2½ inches up to 16 inches are intended for use in naval vessels steam, water, and oil service.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Composition and rating required (see 1.2).

c. C_v required (see 3.3).

d. Standard sizes of drains and by-passes required (see table II).

e. Nonstandard sizes of drains and bypasses required (see 3.3.3.1).

f. Location of drains and bypasses (see 3.3.3.2).

g. Boss requirements and requirement for valve bodies (see 3.3.3.4).

h. Pattern required (see 3.3.4).

i. Port arrangement on globe valves if other than in-line (see 3.3.5).

j. Requirements for a supply list detailing oversized seal rings for repair purposes (see 3.3.9.1.2).

k. Stem configuration if other than rising stem (see 3.3.11.1).

l. Trim materials if other than that specified in table IV (see 3.3.11.5).

m. Specific actuator closure time if different from those of MIL-DTL-24657 (see 3.3.12.6).

n. Allowable Seat leakage if other than specified (see 3.3.13).

o. Requirements for master drawings, unless the drawings had been submitted as part of a previous order (see 3.3.16 and 6.5).

p. AQL of 1.0 (see 4.3.2).

q. Service conditions if not specified herein (see 4.5.1).

r. NDT criteria for valves used in services other than those listed (see table VII).

s. Packaging requirements (see 5.1).

t. Documentation requirements (see 6.4).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List (QPL) No. 22052, whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Documentation. The following is a list of documentation that may be requested as part of the acquisition of any valve under this specification (see 6.2):

a. Engineering drawings.

b. Master drawings.

c. Materials listing.

d. Installation drawings.

e. Technical manual.

f. Test reports.

g. Certifications.

h. On board repair parts list.

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6.5 Subject term (key word) listing.

Bonnet

Bosses

Flow coefficient

Hammer blow

Hard facing

Outside screw

Rising stem

Toggle operator

Valve seat

Yokes

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Navy – SH

Preparing activity:
Navy – SH
(Project 4820-2018-008)

Review activity:
DLA – CC

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